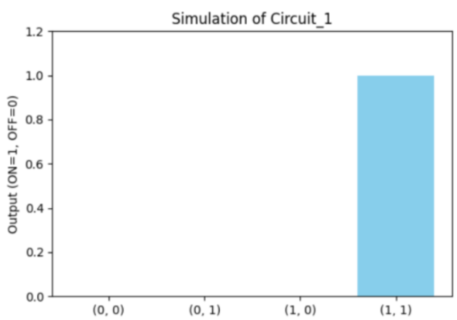
```python
# Example circuits
circuits = {
    "Circuit_1": {"inputs": ["A", "B"], "gate": "AND"},
    "Circuit_2": {"inputs": ["A"], "gate": "NOT"},
    "Circuit_3": {"inputs": ["A", "B"], "gate": "OR"}  # optional extra
}

# Function to simulate a logic gate
def simulate_gate(inputs, gate_type):
    if gate_type == "AND":
        return all(inputs)
    elif gate_type == "OR":
        return any(inputs)
    elif gate_type == "NOT":
        return not inputs[0]
    else:
        raise ValueError("Unknown gate type")

# Test simulation with all input combinations for Circuit 1
from itertools import product

circuit_to_test = "Circuit_1"
inputs = circuits[circuit_to_test]["inputs"]
gate = circuits[circuit_to_test]["gate"]

# Generate all combinations of 0 and 1 for inputs
input_combinations = list(product([0,1], repeat=len(inputs)))
```

```python
# Simulate outputs
simulation_results = []
for combo in input_combinations:
    output = simulate_gate(combo, gate)
    simulation_results.append({"inputs": combo, "output": output})

# Display results
import pandas as pd
df_sim = pd.DataFrame(simulation_results)
print(df_sim)
```

```
   inputs  output
0  (0, 0)   False
1  (0, 1)   False
2  (1, 0)   False
3  (1, 1)    True
```

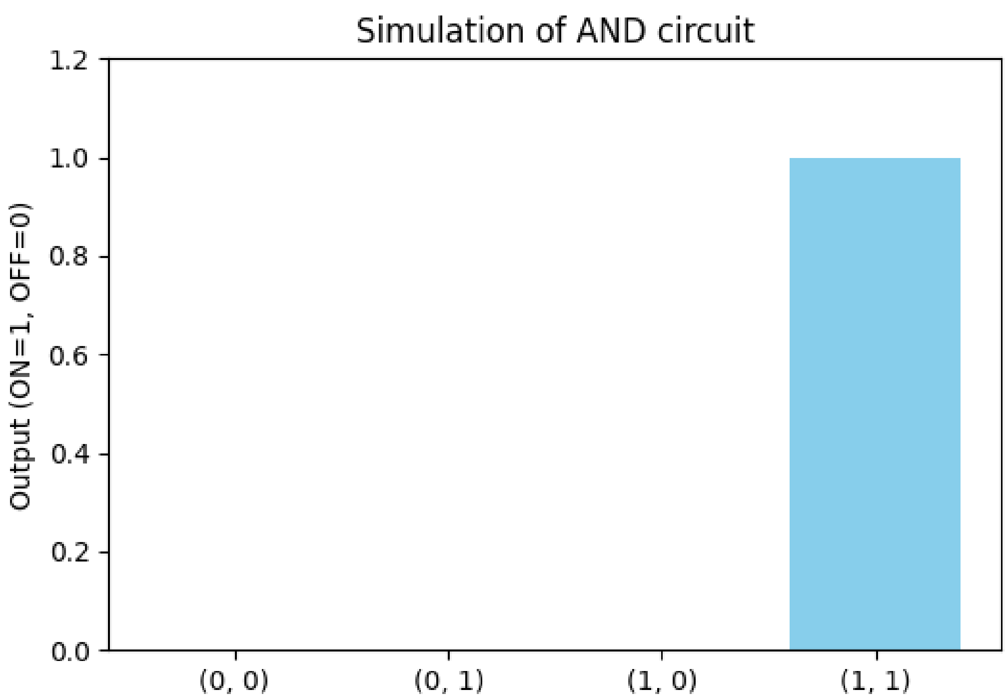
```python
import matplotlib.pyplot as plt

# Prepare data for plotting
x_labels = [str(combo) for combo in df_sim['inputs']]
y_values = [int(val) for val in df_sim['output']]  # 1 for True, 0 for False

# Plot
plt.figure(figsize=(6,4))
plt.bar(x_labels, y_values, color='skyblue')
plt.ylim(0,1.2)
```

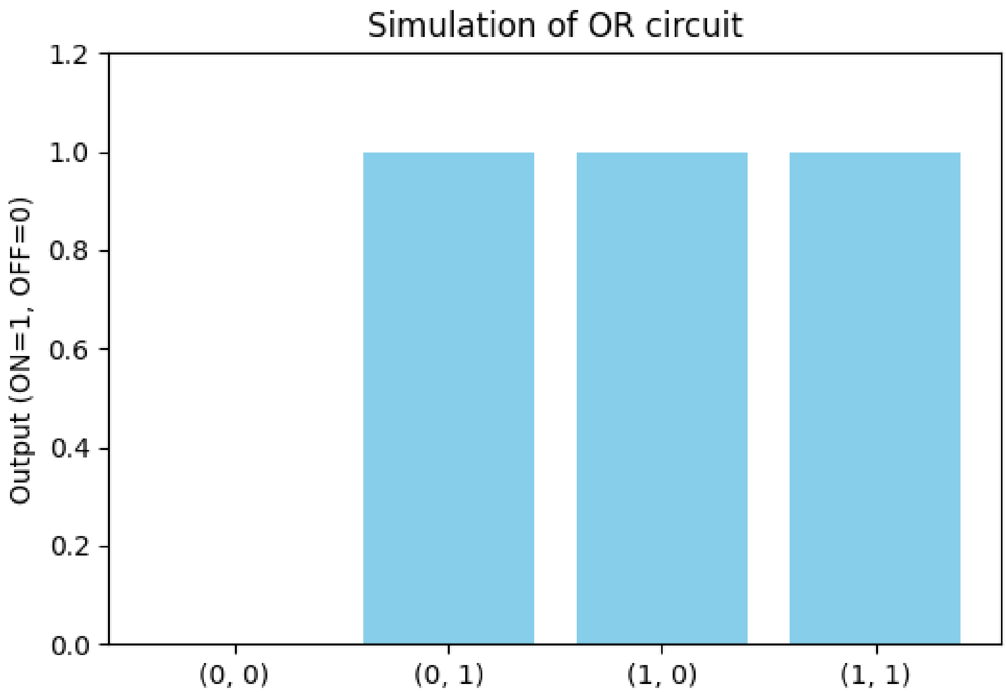
```python
plt.ylabel('Output (ON=1, OFF=0)')
plt.title(f"Simulation of {circuit_to_test}")
plt.show()
```

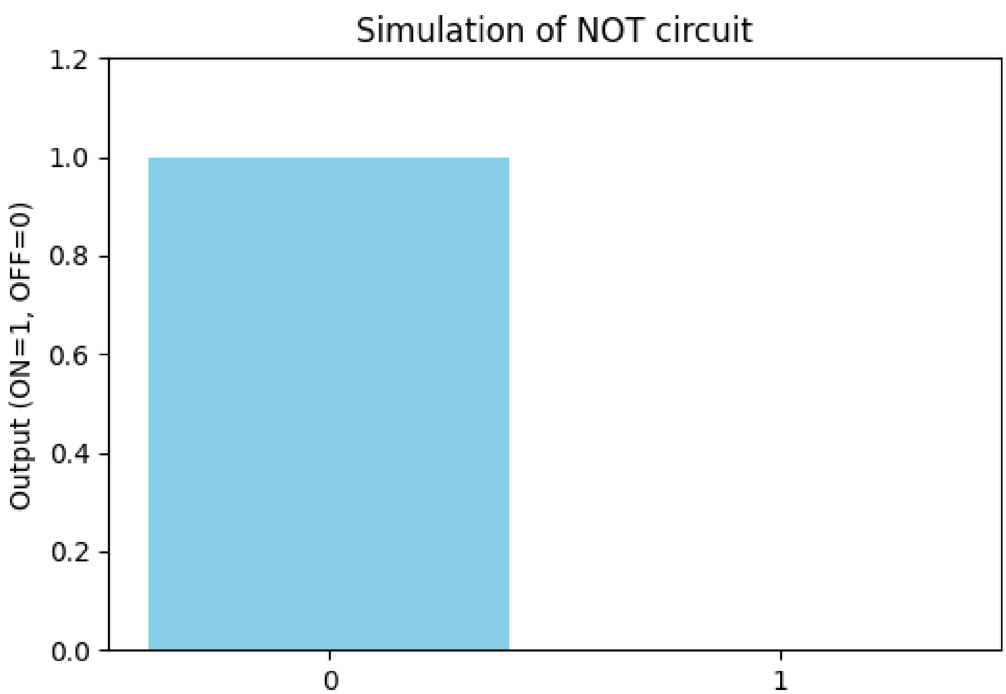


```python
import pandas as pd

# AND gate
df_and = pd.DataFrame({
    "inputs": [(0,0),(0,1),(1,0),(1,1)],
    "output": [False, False, False, True]
})

# OR gate
df_or = pd.DataFrame({
    "inputs": [(0,0),(0,1),(1,0),(1,1)],
    "output": [False, True, True, True]
})

# NOT gate
df_not = pd.DataFrame({
    "inputs": [0, 1],
    "output": [True, False]
})
```

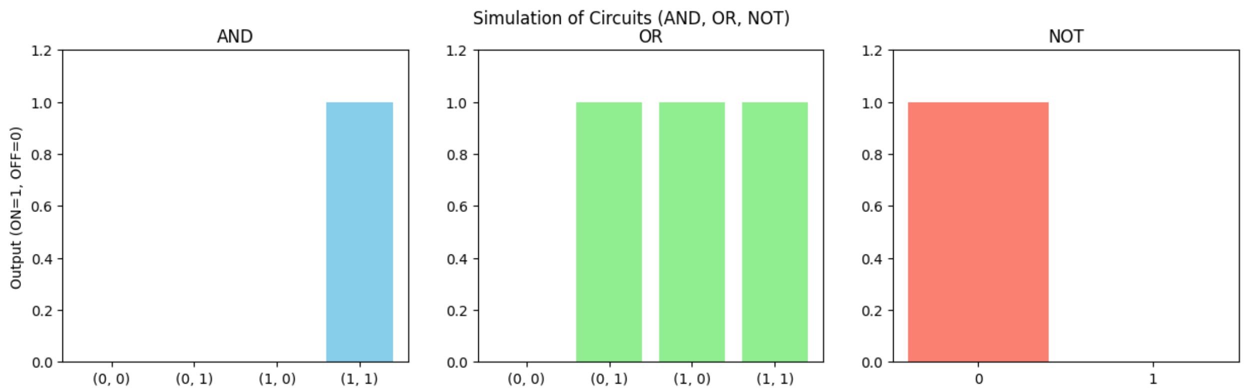
```python
import matplotlib.pyplot as plt

def plot_circuit(df, name):
    x_labels = [str(i) for i in df['inputs']]
    y_values = [int(val) for val in df['output']]
    plt.figure(figsize=(6,4))
    plt.bar(x_labels, y_values, color='skyblue')
    plt.ylim(0,1.2)
    plt.ylabel('Output (ON=1, OFF=0)')
    plt.title(f"Simulation of {name} circuit")
    plt.show()

# Plot all circuits
plot_circuit(df_and, "AND")
plot_circuit(df_or, "OR")
plot_circuit(df_not, "NOT")
```




Simulation of OR circuit
Output (ON=1, OFF=0)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
(0, 0)
(0, 1)
(1, 0)
(1, 1)


Simulation of NOT circuit
Output (ON=1, OFF=0)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0
1

```python
import matplotlib.pyplot as plt

# Combine all circuits into one figure
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

# AND circuit
axes[0].bar([str(i) for i in df_and['inputs']], [int(val) for val in df_and['output']], color='skyblue')
axes[0].set_ylim(0,1.2)
axes[0].set_title("AND")
axes[0].set_ylabel('Output (ON=1, OFF=0)')

# OR circuit
axes[1].bar([str(i) for i in df_or['inputs']], [int(val) for val in df_or['output']], color='lightgreen')
axes[1].set_ylim(0,1.2)
axes[1].set_title("OR")

# NOT circuit
axes[2].bar([str(i) for i in df_not['inputs']], [int(val) for val in df_not['output']], color='salmon')
axes[2].set_ylim(0,1.2)
axes[2].set_title("NOT")

plt.suptitle("Simulation of Circuits (AND, OR, NOT)")
plt.show()
```




```python
import matplotlib.pyplot as plt

# Helper function to add ON/OFF labels
def add_labels(ax, outputs):
    for i, val in enumerate(outputs):
        label = 'ON' if val else 'OFF'
        ax.text(i, val + 0.05, label, ha='center', va='bottom', fontsize=10)

# Combine all circuits into one figure
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

# AND circuit
axes[0].bar([str(i) for i in df_and['inputs']], [int(val) for val in df_and['output']], color='skyblue')
axes[0].set_ylim(0,1.2)
axes[0].set_title("AND")
axes[0].set_ylabel('Output')
add_labels(axes[0], df_and['output'])

# OR circuit
axes[1].bar([str(i) for i in df_or['inputs']], [int(val) for val in df_or['output']], color='lightgreen')
axes[1].set_ylim(0,1.2)
axes[1].set_title("OR")
add_labels(axes[1], df_or['output'])
```

```python
axes[1].set_ylim(0,1.2)
axes[1].set_title("OR")
add_labels(axes[1], df_or['output'])

# NOT circuit
axes[2].bar([str(i) for i in df_not['inputs']], [int(val) for val in df_not['output']], color='salmon')
axes[2].set_ylim(0,1.2)
axes[2].set_title("NOT")
add_labels(axes[2], df_not['output'])

plt.suptitle("Simulation of Circuits (AND, OR, NOT)")
plt.show()
```